# Common Core State Standards Correlations


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# Correlation of *Core Connections, Course 3* to the Common Core State Standards Grade 8

This document identifies the lesson(s) in *Core Connections, Course 3* in which primary instruction of each standard from the Common Core State Standards for Mathematics, Grade 8 occurs.

**The standards continue to be implemented, applied, and practiced throughout subsequent lessons.** *#.#.#* refers to a lesson in *Core Connections, Course 3*. MN *#.#.#* refers to the Math Notes box in Lesson *#.#.#*, Checkpoint # refers to the Checkpoint problems in the back of the student textbook, and #-# refers to an instance of a homework problem in which the standard is implemented. The list of homework problems is by no means comprehensive, but rather intended as a sample.

| Standard from CCSS Grade 8 | Primary instruction for this standard and a few instances of homework in which the standard is implemented. |
|---|---|
| **The Number System 8.NS** | |
| **Know that there are numbers that are not rational, and approximate them by rational numbers.** | |
| 8.NS.1. Know that numbers that are not rational are called irrational. Understand informally that every number has a decimal expansion; for rational numbers show that the decimal expansion repeats eventually, and convert a decimal expansion which repeats eventually into a rational number. | 9.2.4<br>9-110, 9-116, 9-131, 10-12, 10-15, 10-28, 10-67 |
| 8.NS.2. Use rational approximations of irrational numbers to compare the size of irrational numbers, locate them approximately on a number line diagram, and estimate the value of expressions (e.g., $\pi^2$). *For example, by truncating the decimal expansion of* $\sqrt{2}$*, show that* $\sqrt{2}$ *is between 1 and 2, then between 1.4 and 1.5, and explain how to continue on to get better approximations.* | 9.2.3 |

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| **Expressions and Equations 8.EE** | |
| **Work with radicals and integer exponents.** | |
| 8.EE.1. Know and apply the properties of integer exponents to generate equivalent numerical expressions. *For example,* $3^2 \times 3^{-5} = 3^{-3} = 1/3^3 = 1/27$. | 8.2.1–8.2.3<br>8-47, 8-55, 8-66, 8-89, 8-95, 8-110, 8-136, 9-114, 9-133, 9-142, 9-158, 10-29, 10-68 |
| 8.EE.2. Use square root and cube root symbols to represent solutions to equations of the form $x^2 = p$ and $x^3 = p$, where $p$ is a positive rational number. Evaluate square roots of small perfect squares and cube roots of small perfect cubes. Know that $\sqrt{2}$ is irrational. | 9.2.3, 9.2.4, 10.1.1<br>9-89, 9-116, 9-157, 10-12, 10-13, 10-56, 10-77 |



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| 8.EE.3. Use numbers expressed in the form of a single digit times an integer power of 10 to estimate very large or very small quantities, and to express how many times as much one is than the other. *For example, estimate the population of the United States as $3 \times 10^8$ and the population of the world as $7 \times 10^9$, and determine that the world population is more than 20 times larger.* | 8.2.1<br>8-53, 8-54, 8-88, 8-94, 8-138, 9-66, 9-99, 9-113 |
| 8.EE.4. Perform operations with numbers expressed in scientific notation, including problems where both decimal and scientific notation are used. Use scientific notation and choose units of appropriate size for measurements of very large or very small quantities (e.g., use millimeters per year for seafloor spreading). Interpret scientific notation that has been generated by technology. | 8.2.4<br>8-109, 8-125, 8-134, 10-65, 10-76 |
| **Understand the connections between proportional relationships, lines, and linear equations.** | |
| 8.EE.5. Graph proportional relationships, interpreting the unit rate as the slope of the graph. Compare two different proportional relationships represented in different ways. *For example, compare a distance-time graph to a distance-time equation to determine which of two moving objects has greater speed.* | 1.2.1, Checkpoint 3 (3-117), 7.2.5<br>MN: 7.2.4<br>3-67, 3-117, 6-59, 7-75, 8-132 |
| 8.EE.6. Use similar triangles to explain why the slope $m$ is the same between any two distinct points on a non-vertical line in the coordinate plane; derive the equation $y = mx$ for a line through the origin and the equation $y = mx + b$ for a line intercepting the vertical axis at $b$. | 4.1.4, 7.2.1–7.2.5<br>4-38, 4-68, 4-72, 5-29, 7-105, 7-119 |
| **Analyze and solve linear equations and pairs of simultaneous linear equations.** | |
| 8.EE.7. Solve linear equations in one variable. | |
| 8.EE.7a. Give examples of linear equations in one variable with one solution, infinitely many solutions, or no solutions. Show which of these possibilities is the case by successively transforming the given equation into simpler forms, until an equivalent equation of the form $x = a$, $a = a$, or $a = b$ results (where $a$ and $b$ are different numbers). | 2.1.8, 2.1.9, 3.2.1–3.2.2, Checkpoint 5 (5-61)<br>3-94, 3-126, 5-61, 6-4, 6-73 |
| 8.EE.7b. Solve linear equations with rational number coefficients, including equations whose solutions require expanding expressions using the distributive property and collecting like terms. | 2.1.8, 2.1.9, 3.2.1–3.2.5, 5.1.2<br>MN: 5.2.1<br>3-74, 3-113, 4-39, 5-22, 6-83, 6-85, 6-115, 7-33, 7-120 |
| 8.EE.8. Analyze and solve pairs of simultaneous linear equations. | |
| 8.EE.8a. Understand that solutions to a system of two linear equations in two variables correspond to points of intersection of their graphs, because points of intersection satisfy both equations simultaneously. | 5.2.1<br>5-28, 5-36, 5-47, 5-66, 9-48, 9-93 |
| 8.EE.8b. Solve systems of two linear equations in two variables algebraically, and estimate solutions by graphing the equations. Solve simple cases by inspection. *For example, $3x + 2y = 5$ and $3x + 2y = 6$ have no solution because $3x + 2y$ cannot simultaneously be 5 and 6.* | 5.2.3, 5.2.4<br>5-36, 5-48, 5-57, 5-66, 7-10, 9-48, 9-63 |


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| 8.EE.8c. Solve real-world and mathematical problems leading to two linear equations in two variables. *For example, given coordinates for two pairs of points, determine whether the line through the first pair of points intersects the line through the second pair.* | 5.2.2–5.2.4<br>5-36, 5-67, 6-114, 8-131, 9-76, 10-79 |

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| **Functions 8.F** | |
| **Define, evaluate, and compare functions.** | |
| 8.F.1. Understand that a function is a rule that assigns to each input exactly one output. The graph of a function is the set of ordered pairs consisting of an input and the corresponding output. (function notation not required) | 3.1.2–3.1.6, 8.3.1<br>3-125, 4-19, 4-75, 9-13, 9-92, 10-58 |
| 8.F.2. Compare properties of two functions each represented in a different way (algebraically, graphically, numerically in tables, or by verbal descriptions). *For example, given a linear function represented by a table of values and a linear function represented by an algebraic expression, determine which function has the greater rate of change.* | 3.1.2–3.1.6, 4.1.1–4.1.7, Checkpoint 6 (6-109), 7.2.1<br>5-GW (p. 217), 6-32, 6-109 |
| 8.F.3. Interpret the equation $y = mx + b$ as defining a linear function, whose graph is a straight line; give examples of functions that are not linear. *For example, the function $A = s^2$ giving the area of a square as a function of its side length is not linear because its graph contains the points (1,1), (2,4) and (3,9), which are not on a straight line.* | 3.1.2–3.1.6, 7.2.1, 8.1.1, 8.1.3<br>3-76, 3-97, 3-105, 4-75, 6-41, 8-21, 8-58, 9-60 |
| **Use functions to model relationships between quantities.** | |
| 8.F.4. Construct a function to model a linear relationship between two quantities. Determine the rate of change and initial value of the function from a description of a relationship or from two $(x, y)$ values, including reading these from a table or from a graph. Interpret the rate of change and initial value of a linear function in terms of the situation it models, and in terms of its graph or a table of values. | 3.1.2–3.1.7, 4.1.1–4.1.7<br>4-25, 4-37, 5-296-51, 6-72, 8-127, 9-64 |
| 8.F.5. Describe qualitatively the functional relationship between two quantities by analyzing a graph (e.g., where the function is increasing or decreasing, linear or nonlinear). Sketch a graph that exhibits the qualitative features of a function that has been described verbally. | 8.3.1<br>1-50, 1-68, 4-53, 7-51, 9-46, 10-59, 10-71 |

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| **Geometry 8.G** | |
| **Understand congruence and similarity using physical models, transparencies, or geometry software.** | |
| 8.G.1. Verify experimentally the properties of rotations, reflections, and translations: | |
| 8.G.1a. Lines are taken to lines, and line segments to line segments of the same length. | 6.1.1–6.1.3, 6.2.3<br>6-28, 6-36, 6-106, 6-112, 7-11, 7-83 |


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|---|---|
| 8.G.1b. Angles are taken to angles of the same measure. | 6.1.1–6.1.3, 6.2.3<br>6-28, 6-36, 6-106, 6-112, 7-11, 7-83 |
| 8.G.1c. Parallel lines are taken to parallel lines. | 6.1.1–6.1.3, 6.2.3<br>6-28, 6-36, 6-106, 7-11, 7-83 |
| 8.G.2. Understand that a two-dimensional figure is congruent to another if the second can be obtained from the first by a sequence of rotations, reflections, and translations; given two congruent figures, describe a sequence that exhibits the congruence between them. | 6.1.2–6.1.3, 6.2.3–6.2.4<br>6-28, 6-80 |
| 8.G.3. Describe the effect of dilations, translations, rotations, and reflections on two-dimensional figures using coordinates. | 6.1.2–6.1.3, 6.2.1–6.2.2, Checkpoint 8 (8-135)<br>6-13, 6-27, 6-36, 6-46, 8-34, 8-69, 8-98, 8-135 |
| 8.G.4. Understand that a two-dimensional figure is similar to another if the second can be obtained from the first by a sequence of rotations, reflections, translations, and dilations; given two similar two-dimensional figures, describe a sequence that exhibits the similarity between them. | 6.1.2–6.1.3, 6.2.2–6.2.4<br>6-58, 6-92, 7-34 |
| 8.G.5. Use informal arguments to establish facts about the angle sum and exterior angle of triangles, about the angles created when parallel lines are cut by a transversal, and the angle-angle criterion for similarity of triangles. *For example, arrange three copies of the same triangle so that the sum of the three angles appears to form a line, and give an argument in terms of transversals why this is so.* | 9.1.1–9.1.4<br>9-8, 9-9, 9-21, 9-22, 9-33, 9-34, 9-45, 9-49, 9-56, 9-78, 9-155, 9-156, 10-61 |
| **Understand and apply the Pythagorean Theorem.** | |
| 8.G.6. Explain a proof of the Pythagorean Theorem and its converse. | 9.2.2, 9.2.7 |
| 8.G.7. Apply the Pythagorean Theorem to determine unknown side lengths in right triangles in real-world and mathematical problems in two and three dimensions. | 9.2.2, 9.2.5, 9.2.6<br>9-130, 9-139, 9-140, 9-144, 9-157, 10-14, 10-28 |
| 8.G.8. Apply the Pythagorean Theorem to find the distance between two points in a coordinate system. | 9.2.5<br>9-110, 9-129, 9-149, 10-67 |
| **Solve real-world and mathematical problems involving volume of cylinders, cones, and spheres.** | |
| 8.G.9. Know the formulas for the volumes of cones, cylinders, and spheres and use them to solve real-world and mathematical problems. | 10.1.2–10.1.5<br>10-26, 10-41, 10-42, 10-55, 10-56, 10-65, 10-66, 10-73, 10-74 |


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| **Statistics and Probability 8.SP** | |
|---|---|
| **Investigate patterns of association in bivariate data.** | |
| 8.SP.1. Construct and interpret scatter plots for bivariate measurement data to investigate patterns of association between two quantities. Describe patterns such as clustering, outliers, positive or negative association, linear association, and nonlinear association. | 7.1.2, 7.1.3, Checkpoint 9 (9-50)<br>7-17, 7-29, 7-72, 7-102, 7-103, 8-111, 9-10, 9-24, 9-50, 9-162 |
| 8.SP.2. Know that straight lines are widely used to model relationships between two quantitative variables. For scatter plots that suggest a linear association, informally fit a straight line, and informally assess the model fit by judging the closeness of the data points to the line. | 7.1.3, 7.3.2<br>7-30, 7-91, 7-102, 7-103, 7-112, 8-113, 9-50, 9-162 |
| 8.SP.3. Use the equation of a linear model to solve problems in the context of bivariate measurement data, interpreting the slope and intercept. *For example, in a linear model for a biology experiment, interpret a slope of 1.5 cm/hr as meaning that an additional hour of sunlight each day is associated with an additional 1.5 cm in mature plant height.* | 7.3.1, 7.3.2<br>7-30, 7-103, 9-50 |
| 8.SP.4. Understand that patterns of association can also be seen in bivariate categorical data by displaying frequencies and relative frequencies in a two-way table. Construct and interpret a two-way table summarizing data on two categorical variables collected from the same subjects. Use relative frequencies calculated for rows or columns to describe possible association between the two variables. *For example, collect data from students in your class on whether or not they have a curfew on school nights and whether or not they have assigned chores at home. Is there evidence that those who have a curfew also tend to have chores?* | 7.3.3<br>7-110, 7-111, 8-11, 8-50, 8-97 |



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# Correlation of the Common Core State Standards Grade 8 to *Core Connections, Course 3*

## Chapter 1 Teacher Guide

| Section | Lesson | Lesson Objectives | CCSS Standards |
|---|---|---|---|
| 1.1 | 1.1.1 | Interpreting Graphs | |
| | 1.1.2 | Finding and Generalizing Patterns | |
| | 1.1.3 | The Algebra Walk | |
| | 1.1.4 | Collecting, Organizing, and Analyzing Data | 8.SP.2 |
| 1.2 | 1.2.1 | Proportional Relationships with Graphs and Tables | 8.EE.5 |
| | 1.2.2 | Strategies for Solving Proportional Relationships | Preparation for 8.EE.7a, 8.EE.7b in Section 3.2 |

## Chapter 2 Teacher Guide

| Section | Lesson | Lesson Objectives | CCSS Standards |
|---|---|---|---|
| 2.1 | 2.1.1 | Exploring Variables and Expressions | Preparation for 8.EE.7a, 8.EE.7b in Lesson 2.1.8 |
| | 2.1.2 | Simplifying Expressions by Combining Like Terms | Preparation for 8.EE.7a, 8.EE.7b in Lesson 2.1.8 |
| | 2.1.3 | Writing Algebraic Expressions | Preparation for 8.EE.7a, 8.EE.7b in Lesson 2.1.8 |
| | 2.1.4 | Using Zero to Simplify Algebraic Expressions | Preparation for 8.EE.7a, 8.EE.7b in Lesson 2.1.8 |
| | 2.1.5 | Using Algebra Tiles to Simplify Algebraic Expressions | Preparation for 8.EE.7a, 8.EE.7b in Lesson 2.1.8 |
| | 2.1.6 | Using Algebra Tiles to Compare Expressions | Preparation for 8.EE.7a, 8.EE.7b in Lesson 2.1.8 |
| | 2.1.7 | Simplifying and Recording Work | Preparation for 8.EE.7a, 8.EE.7b in Lesson 2.1.8 |
| | 2.1.8 | Using Algebra Tiles to Solve for $x$ | 8.EE.7a, 8.EE.7b |
| | 2.1.9 | More Solving Equations | 8.EE.7a, 8.EE.7b |



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## Chapter 3 Teacher Guide

| Section | Lesson | Lesson Objectives | CCSS Standards |
|---|---|---|---|
| 3.1 | 3.1.1 | Extending Patterns and Finding Rules | Preparation for 8.F.1 in Lesson 3.1.2 |
| | 3.1.2 | Using Tables, Graphs, and Rules to Make Predictions | 8.F.1, 8.F.2, 8.F.3, 8.F.4 |
| | 3.1.3 | Using a Graphing Calculator and Identifying Solutions | 8.F.1, 8.F.2, 8.F.3, 8.F.4 |
| | 3.1.4 | Completing Tables and Drawing Graphs | 8.F.1, 8.F.2, 8.F.3, 8.F.4 |
| | 3.1.5 | Graphs, Tables, and Rules | 8.F.1, 8.F.2, 8.F.3, 8.F.4 |
| | 3.1.6 | Complete Graphs | 8.F.1, 8.F.2, 8.F.3, 8.F.4 |
| | 3.1.7 | Identifying Common Graphing Errors | 8.F.4 |
| 3.2 | 3.2.1 | Solving Equations and Checking Solutions | 8.EE.7a, 8.EE.7b |
| | 3.2.2 | Determining the Number of Solutions | 8.EE.7a, 8.EE.7b |
| | 3.2.3 | Problem Solving With Equations | 8.EE.7b |
| | 3.2.4 | More Solving Equations to Solve Problems | 8.EE.7b |
| | 3.2.5 | Distributive Property Equations | 8.EE.7b |


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## Chapter 4 Teacher Guide

| Section | Lesson | Lesson Objectives | CCSS Standards |
|---|---|---|---|
| 4.1 | 4.1.1 | Finding Connections Between Representations | 8.F.2, 8.F.4 |
| | 4.1.2 | Seeing Growth in Different Representations | 8.F.2, 8.F.4 |
| | 4.1.3 | Connecting Linear Rules and Graphs | 8.F.2, 8.F.4 |
| | 4.1.4 | $y = mx + b$ | 8.EE.6, 8.F.2, 8.F.4 |
| | 4.1.5 | Checking the Connections | 8.F.2, 8.F.4 |
| | 4.1.6 | Graphing a Line Without an $x \rightarrow y$ Table | 8.F.2, 8.F.4 |
| | 4.1.7 | Completing the Web | 8.F.2, 8.F.4 |

## Chapter 5 Teacher Guide

| Section | Lesson | Lesson Objectives | CCSS Standards |
|---|---|---|---|
| 5.1 | 5.1.1 | Working with Multi-Variable Equations | Preparation for 8.EE.8b in Lesson 5.2.4 |
| | 5.1.2 | Solving Equations with Fractions | 8.EE.7b |
| 5.2 | 5.2.1 | Introduction to Systems of Equations | 8.EE.8a |
| | 5.2.2 | Writing Rules from Word Problems | 8.EE.8c |
| | 5.2.3 | Solving Systems Algebraically | 8.EE.8b, 8.EE.8c |
| | 5.2.4 | Strategies for Solving Systems | 8.EE.8b, 8.EE.8c |


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## Chapter 6 Teacher Guide

| Section | Lesson | Lesson Objectives | CCSS Standards |
|---|---|---|---|
| 6.1 | 6.1.1 | Rigid Transformations | 8.G.1a, 8.G.1b, 8.G.1c |
| | 6.1.2 | Rigid Transformations on a Coordinate Graph | 8.G.1a, 8.G.1b, 8.G.1c, 8.G.2, 8.G.3, 8.G.4 |
| | 6.1.3 | Describing Transformations | 8.G.1a, 8.G.1b, 8.G.1c, 8.G.2, 8.G.3, 8.G.4 |
| | 6.1.4 | Using Rigid Transformations | 8.G.3 |
| 6.2 | 6.2.1 | Multiplication and Dilation | 8.G.3 |
| | 6.2.2 | Dilations and Similar Figures | 8.G.3, 8.G.4 |
| | 6.2.3 | Identifying Similar Shapes | 8.G.1a, 8.G.1b, 8.G.1c, 8.G.2, 8.G.4 |
| | 6.2.4 | Similar Figures and Transformations | 8.G.2, 8.G.4 |
| | 6.2.5 | Working With Corresponding Sides | 8.G.4 |
| | 6.2.6 | Solving Problems Involving Similar Shapes | 8.G.4 |


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## Chapter 7 Teacher Guide

| Section | Lesson | Lesson Objectives | CCSS Standards |
|---|---|---|---|
| 7.1 | 7.1.1 | Circle Graphs | |
| | 7.1.2 | Organizing Data in a Scatterplot | 8.SP.1 |
| | 7.1.3 | Identifying and Describing Association | 8.SP.1, 8.SP.2 |
| 7.2 | 7.2.1 | $y = mx + b$ Revisited | 8.EE.6, 8.F.3 |
| | 7.2.2 | Slope | 8.EE.6 |
| | 7.2.3 | Slope in Different Representations | 8.EE.6 |
| | 7.2.4 | More About Slope | 8.EE.6 |
| | 7.2.5 | Proportional Equations | 8.EE.5, 8.EE.6 |
| 7.3 | 7.3.1 | Using Equations to Make Predictions | 8.SP.3 |
| | 7.3.2 | Describing Association Fully | 8.SP.2, 8.SP.3 |
| | 7.3.3 | Association Between Categorical Variables | 8.SP.4 |

## Chapter 8 Teacher Guide

| Section | Lesson | Lesson Objectives | CCSS Standards |
|---|---|---|---|
| 8.1 | 8.1.1 | Patterns of Growth in Tables and Graphs | 8.F.3 |
| | 8.1.2 | Compound Interest | Preparation for 8.EE.1 in Lesson 8.2.1 |
| | 8.1.3 | Linear and Exponential Growth | 8.F.3 |
| 8.2 | 8.2.1 | Exponents and Scientific Notation | 8.EE.1, 8.EE.3 |
| | 8.2.2 | Exponent Rules | 8.EE.1 |
| | 8.2.3 | Negative Exponents | 8.EE.1 |
| | 8.2.4 | Operations with Scientific Notation | 8.EE.4 |
| 8.3 | 8.3.1 | Functions in Graphs and Tables | 8.F.1, 8.F.3, 8.F.5 |


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## Chapter 9 Teacher Guide

| Section | Lesson | Lesson Objectives | CCSS Standards |
|---|---|---|---|
| 9.1 | 9.1.1 | Parallel Line Angle Pair Relationships | 8.G.5 |
| | 9.1.2 | Finding Unknown Angles in Triangles | 8.G.5 |
| | 9.1.3 | Exterior Angles in Triangles | 8.G.5 |
| | 9.1.4 | AA Triangle Similarity | 8.G.5 |
| 9.2 | 9.2.1 | Side Lengths and Triangles | Preparation for 8.G.7 in Lesson 9.2.2 |
| | 9.2.2 | Pythagorean Theorem | 8.G.6, 8.G.7 |
| | 9.2.3 | Understanding Square Root | 8.NS.2 |
| | 9.2.4 | Real Numbers | 8.NS.1, 8.EE.2 |
| | 9.2.5 | Applications of Pythagorean Theorem | 8.G.7, 8.G.8 |
| | 9.2.6 | Pythagorean Theorem in Three Dimensions | 8.G.7 |
| | 9.2.7 | Pythagorean Theorem Proofs | 8.G.6 |

## Chapter 10 Teacher Guide

| Section | Lesson | Lesson Objectives | CCSS Standards |
|---|---|---|---|
| 10.1 | 10.1.1 | Cube Roots | 8.EE.2 |
| | 10.1.2 | Surface Area and Volume of a Cylinder | 8.G.9 |
| | 10.1.3 | Volumes of Cones and Pyramids | 8.G.9 |
| | 10.1.4 | Volume of a Sphere | 8.G.9 |
| | 10.1.5 | Applications of Volume | 8.G.9 |


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# CCSS Standards for Mathematical Practice in CPM Courses

All lessons in a CPM course integrate the CCSS Standards for Mathematical Practice. CPM does not treat the CCSS Standards for Mathematical Practice as an occasional activity, nor does it simply tack them onto the lessons. The CCSS Standards for Mathematical Practice follow, along with descriptions of how they are implemented throughout the CPM curriculum.

**1. Make sense of problems and persevere in solving them.**
Mathematically proficient students start by explaining to themselves the meaning of a problem and looking for entry points to its solution. They analyze givens, constraints, relationships, and goals. They make conjectures about the form and meaning of the solution and plan a solution pathway rather than simply jumping into a solution attempt. They consider analogous problems, and try special cases and simpler forms of the original problem in order to gain insight into its solution. They monitor and evaluate their progress and change course if necessary. Older students might, depending on the context of the problem, transform algebraic expressions or change the viewing window on their graphing calculator to get the information they need. Mathematically proficient students can explain correspondences between equations, verbal descriptions, tables, and graphs or draw diagrams of important features and relationships, graph data, and search for regularity or trends. Younger students might rely on using concrete objects or pictures to help conceptualize and solve a problem. Mathematically proficient students check their answers to problems using a different method, and they continually ask themselves, "Does this make sense?" They can understand the approaches of others to solving complex problems and identify correspondences between different approaches.

The CPM courses have students solve realistic, non-routine problems that are rich in mathematics on a daily basis. The guided investigations are not merely a set of "word problems" that mimic algorithmic, computational problems. By having students make sense of each problem, rather than being told how to solve a particular kind of problem step by step, the CPM curriculum helps students to develop deep conceptual understanding of the mathematics, procedural fluency, and perseverance. In addition, the curriculum teaches problem-solving strategies that equip students with strategic competence and adaptive reasoning.

**2. Reason abstractly and quantitatively.**
Mathematically proficient students make sense of quantities and their relationships in problem situations. They bring two complementary abilities to bear on problems involving quantitative relationships: the ability to *decontextualize*—to abstract a given situation and represent it symbolically and manipulate the representing symbols as if they have a life of their own, without necessarily attending to their referents—and the ability to *contextualize*, to pause as needed during the manipulation process in order to probe into the referents for the symbols involved. Quantitative reasoning entails habits of creating a coherent representation of the problem at hand; considering the units involved; attending to the meaning of quantities, not just how to compute them; and knowing and flexibly using different properties of operations and objects.

Instead of offering word problems only at the end of each chapter or problem set, the CPM curriculum generally presents mathematical ideas in contexts *first*, helping students make sense of otherwise abstract principles. Only then do students move on to abstraction and generalization using symbolic notation. Students are taught to try specific numbers to evaluate what is happening in problems, critically analyze units, identify what numbers or solutions mean in a given context, write number sentences or expressions that represent given relationships, and identify patterns and relationships that lead to solutions.



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**3. Construct viable arguments and critique the reasoning of others.**
Mathematically proficient students understand and use stated assumptions, definitions, and previously established results in constructing arguments. They make conjectures and build a logical progression of statements to explore the truth of their conjectures. They are able to analyze situations by breaking them into cases, and can recognize and use counterexamples. They justify their conclusions, communicate them to others, and respond to the arguments of others. They reason inductively about data, making plausible arguments that take into account the context from which the data arose. Mathematically proficient students are also able to compare the effectiveness of two plausible arguments, distinguish correct logic or reasoning from that which is flawed, and—if there is a flaw in an argument—explain what it is. Elementary students can construct arguments using concrete referents such as objects, drawings, diagrams, and actions. Such arguments can make sense and be correct, even though they are not generalized or made formal until later grades. Later, students learn to determine domains to which an argument applies. Students at all grades can listen or read the arguments of others, decide whether they make sense, and ask useful questions to clarify or improve the arguments.

In CPM courses, students regularly share information, opinions, and their knowledge and understandings in study teams. They contribute, listen, argue, ask each other for help, check for understanding, and keep each other focused. During this process students use higher-order thinking skills by providing clarification, building on the ideas of others, analyzing and coming to consensus. Justifying and critiquing happens every day in a CPM classroom; it does not occur as an occasional assignment. Students must communicate their mathematical findings in a clear and convincing manner, via narratives, oral presentations, or posters. Teachers answer students' questions, doing so in a way that challenges and motivates students to conjecture and test solutions themselves.

**4. Model with mathematics.**
Mathematically proficient students can apply the mathematics they know to solve problems arising in everyday life, society, and the workplace. In early grades, this might be as simple as writing an addition equation to describe a situation. In middle grades, a student might apply proportional reasoning to plan a school event or analyze a problem in the community. By high school, a student might use geometry to solve a design problem or use a function to describe how one quantity of interest depends on another. Mathematically proficient students who can apply what they know are comfortable making assumptions and approximations to simplify a complicated situation, realizing that these may need revision later. They are able to identify important quantities in a practical situation and map their relationships using such tools as diagrams, two-way tables, graphs, flowcharts and formulas. They can analyze those relationships mathematically to draw conclusions. They routinely interpret their mathematical results in the context of the situation and reflect on whether the results make sense, possibly improving the model if it has not served its purpose.

Modeling contextual situations with multiple representations recurs throughout the CPM curriculum. From their first encounters with basic functions like proportions and linear equations, students learn to model situations using tables, graphs, equations, written descriptions, and diagrams. This approach continues throughout each course, but with increasingly complex functions. In doing so, students make assumptions, form predictions, and then check to see if their predictions make sense in the context of the problem. Students use area models to multiply fractions, multiply and divide polynomials, factor, and solve probability problems. Students learn that models may not be perfect, yet can be very useful for describing data and making predictions.


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**5. Use appropriate tools strategically.**
Mathematically proficient students consider the available tools when solving a mathematical problem. These tools might include pencil and paper, concrete models, a ruler, a protractor, a calculator, a spreadsheet, a computer algebra system, a statistical package, or dynamic geometry software. Proficient students are sufficiently familiar with tools appropriate for their grade or course to make sound decisions about when each of these tools might be helpful, recognizing both the insight to be gained and their limitations. For example, mathematically proficient high school students analyze graphs of functions and solutions generated using a graphing calculator. They detect possible errors by strategically using estimation and other mathematical knowledge. When making mathematical models, they know that technology can enable them to visualize the results of varying assumptions, explore consequences, and compare predictions with data. Mathematically proficient students at various grade levels are able to identify relevant external mathematical resources, such as digital content located on a website, and use them to pose or solve problems. They are able to use technological tools to explore and deepen their understanding of concepts.

In a typical CPM lesson, students have the option of using various tools—including constructing physical models, drawing diagrams, using manipulatives, using calculators, or using eTools—but are not typically told which specific tools to use to solve any particular problem. It is not unusual for different teams to use different tools to solve a problem. During a typical lesson, students share their solution strategies with the whole class, which frequently provokes a lively discussion of which tools were most efficient and productive to solve a given problem.

**6. Attend to precision.**
Mathematically proficient students try to communicate precisely to others. They try to use clear definitions in discussion with others and in their own reasoning. They state the meaning of the symbols they choose, including using the equal sign consistently and appropriately. They are careful about specifying units of measure, and labeling axes to clarify the correspondence with quantities in a problem. They calculate accurately and efficiently, express numerical answers with a degree of precision appropriate for the problem context. In the elementary grades, students give carefully formulated explanations to each other. By the time they reach high school they have learned to examine claims and make explicit use of definitions.

When using the CPM curriculum, students must communicate clearly, giving carefully formulated explanations to each other, hence they understand the need for attending to precision. While working on problems, students pay attention to units and check whether a numerical solution makes sense. Other times they are precise in scaling or reading the scale on a graph. The use of calculators in some CPM investigations requires students to attend to the precision of the results displayed.


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**7. Look for and make use of structure.**
Mathematically proficient students look closely to discern a pattern or structure. Young students, for example, might notice that three and seven more is the same amount as seven and three more, or they may sort a collection of shapes according to how many sides the shapes have. Later, students will see $7 \times 8$ equals the well remembered $7 \times 5 + 7 \times 3$, in preparation for learning about the distributive property. In the expression $x^2 + 9x + 14$, older students can see the 14 as $2 \times 7$ and the 9 as $2 + 7$. They recognize the significance of an existing line in a geometric figure and can use the strategy of drawing an auxiliary line for solving problems. They also can step back for an overview and shift perspective. They can see complicated things, such as some algebraic expressions, as single objects or as being composed of several objects. For example, they can see $5 - 3(x - y)^2$ as 5 minus a positive number times a square and use that to realize that its value cannot be more than 5 for any real numbers $x$ and $y$.

Since CPM students are developing conceptual understanding of the underlying mathematics, they frequently look for and make use of structure. For example, area models are used to understand multiplication and division, or factoring problems. Giant Ones—fractions where the numerator and denominator are equal—are used to simplify rational expressions, using the underlying structure of the expression instead of just 'canceling out.' In the middle school courses, the 5-D Process is used to understand the structure of the expression needed to solve a problem. In the topic of geometry, students use the structures of polygons to classify them and later prove theorems about them.

**8. Look for and express regularity in repeated reasoning.**
Mathematically proficient students notice if calculations are repeated, and look both for general methods and for shortcuts. Upper elementary students might notice when dividing 25 by 11 that they are repeating the same calculations over and over again, and conclude they have a repeating decimal. By paying attention to the calculation of slope as they repeatedly check whether points are on the line through (1, 2) with slope 3, middle school students might abstract the equation $(y - 2)/(x - 1) = 3$. Noticing the regularity in the way terms cancel when expanding $(x - 1)(x + 1)$, $(x - 1)(x^2 + x + 1)$, and $(x - 1)(x^3 + x^2 + x + 1)$ might lead them to the general formula for the sum of a geometric series. As they work to solve a problem, mathematically proficient students maintain oversight of the process, while attending to the details. They continually evaluate the reasonableness of their intermediate results.

In a CPM course, students often look for and generalize patterns. Patterns are used repeatedly to recognize generalizations or make conjectures about rules. Repeated reasoning allows for increasingly complex problems to be solved from simpler ones. For example, students use patterns to make sense of negative, zero, and fractional exponents. They also recognize patterns when factoring different expressions.


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## Integration of the CCSS Mathematical Practices in a Sample Lesson

A typical CPM lesson reveals depth of integration of the CCSS Mathematical Practices into the course. In Lesson 3.1.2 in CPM's *Core Connections, Course 3*, students work in collaborative teams conducting an inquiry into multiple representations of a linear function. In problem 3-9, "John's Giant Redwood," students try to find a rule that represents an input-output function table of haphazard order. They do this in an interactive (and fun!) whole-class game that is set up in such a way that when one student finds a solution it does not end the process for the rest of the students. During the game, students need to make sense of the pattern they see, look for additional patterns and structure, and look for consistency in repeated input-output combinations (Practices 1, 7, and 8). In problem 3-10 students reason quantitatively and abstractly (Practice 2) to justify (Practice 3) their finding of the growth rate and their prediction. Students have calculators readily available, but quickly determine that repeated addition on the calculator is not a very efficient process for solving this problem (Practice 5). In problem 3-11, students model (Practice 4) the growth of the tree with a graph, and start an inquiry into the capabilities of their model; in part (d) of problem 3-11 students justify (Practice 3) predictions made with the model. Problem 3-12 moves students back and forth between the concrete and abstract (Practice 2) to create an equation that models the situation (Practice 4). Students apply the same skills they used in problem 3-9 from the beginning of the day in their interactive game. In the "Closure" activity for the day (as described in the teacher notes), students discuss the advantages and disadvantages of each of the three representations of the growth function, during which the precision of answers (Practice 6) is discussed. The purpose of the lesson is to begin developing a deep conceptual understanding of rate of growth. Students make sense of growth (Practice 1) by constructing arguments and defending them in collaborative groups, and—toward the end of the day—doing so in a whole-class discussion (Practice 3).



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# Correlation of *Core Connections, Course 3* to the CCSS Mathematical Practices

The table below lists sample lessons that the reader can review to see how CCSS Mathematical Practices are embedded in all the lessons. The reader should examine the detailed sample lesson on the previous page before going through any of the lessons on the next page. The table is by no means exhaustive; it illustrates only how the practices are integrated into a few typical lessons. The CCSS Mathematical Practices stand as the foundation of *each* of the daily lessons. An "xx" in the table represents a practice that is a focus of the lesson. An "x" represents a practice that is present in the lesson.

In addition to the daily execution of the CCSS Mathematical Practices as shown in the table below, culminating problems at the end of the course ask students to specifically reflect on when, where, and how they used the CCSS Mathematical Practices during the course (problems 10-83, 10-87, 10-90).

| | **Mathematical Practice** | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| **Lesson #.#.# Title** | **1** | **2** | **3** | **4** | **5** | **6** | **7** | **8** |
| 2.1.4 Using Zero to Simplify Algebraic Expressions | x | | | | | | xx | |
| 2.1.7 Simplifying and Recording Work | | xx | | | | x | | |
| 3.1.7 Identifying Common Graphing Errors | | xx | x | x | | | | |
| 3.2.2 Determining the Number of Solutions | | xx | x | | | x | | |
| 4.1.2 Seeing Growth in Different Representations | | xx | | | | | x | |
| 4.1.5 Checking the Connections | x | | | | xx | | | |
| 5.2.2 Writing Rules from Word Problems | | x | | x | xx | x | | |
| 5.3 Mid-Course Reflection Activities | | x | xx | xx | x | x | x | |
| 6.1.1 Rigid Transformations | xx | | | | x | | | |
| 6.2.2 Dilations and Similar Figures | xx | | x | | | | | |
| 7.1.3 Identifying and Describing Association | | | x | xx | | x | | xx |
| 7.2.3 Slope in Different Representations | | x | x | | | x | | xx |
| 7.3.2 Describing Association Fully | xx | | | x | | x | | |
| 8.2.1 Exponents and Scientific Notation | | | | | | | x | xx |
| 8.3.1 Functions in Graphs and Tables | | | | | | x | xx | |
| 9.1.3 Exterior Angles in Triangles | | xx | | | | x | x | |
| 9.2.4 Real Numbers | | | | | | xx | x | x |
| 10.1.4 Volume of a Sphere | xx | x | | | | | | |



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